

Candidate Name

Centre Number

Candidate Number



ZIMBABWE SCHOOL EXAMINATIONS COUNCIL
General Certificate of Education Ordinary Level

MATHEMATICS
PAPER 1

4004/1
2 hours 30 minutes

SPECIMEN PAPER

Additional materials:
Mathematical Instruments

INSTRUCTIONS TO CANDIDATES

Write your Name, Centre number and Candidate number in the spaces at the top of each page.
Check that all the pages are in the booklet and ask the invigilator for a replacement if there are duplicate or missing pages.

If working is needed for any question, it must be shown in the space below that question.

Omission of essential working will result in loss of marks.

Decimal answers which are not exact should be given correct to three significant figures unless stated otherwise.

INFORMATION FOR CANDIDATES

The number of marks is given in brackets [] at the end of each question or part question.
This paper is marked out of **100**.

This specimen paper consists of 22 printed pages.

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[Turn over

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Answer **all** questions.**NEITHER MATHEMATICAL TABLES NOR SLIDE RULES****NOR CALCULATORS MAY BE USED IN THIS PAPER.****1** Simplify,

(a) $-10 + 7 - 3,$

Answer (a) _____ [1]

(b) $\sqrt{64} - 5^2 .$

Answer (b) _____ [2]

2 Round off 13,519 correct to**(a)** the nearest tenth,

Answer (a) _____ [1]

(b) 2 decimal places,

Answer (b) _____ [1]

(c) 2 significant figures.

Answer (c) _____ [1]

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- 3** (a) Complete the statement below by placing $\in$ or $\notin$ in place of $*$.

$$15 * \{3; 6; 9 \dots 18\}$$

Answer (a) _____ [1]

- (b)** $A = \{x: 1 < x < 10, x \text{ is a prime number}\}$

List all the elements of A.

Answer (b) _____ [2]

- 4** Simplify,

(a) $x^7 \times x^{-3},$

Answer (a) _____ [1]

(b) $(2a^5)^2.$

Answer (b) _____ [2]

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- 5** **(a)** Find the number of days in 8 weeks.

Answer (a) _____ [1]

- (b)** Express 100111_2 as a number in base 10.

Answer(b) _____ [2]

- 6** Farai invested \$500 at the rate of 15% per year simple interest for 5 years.
Calculate the total simple interest he gets after 5 years,

Answer: _____ [3]

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- 7 Write down the next 2 numbers in the given number pattern.

1; 3; 7; 13; 21; ____; ____;

Answer: _____ [3]

- 8 A trader bought 100 cabbages from a farmer at a cost of 50c each and sold them at 80c each.

Calculate the

- (a) profit the trader made,

Answer (a) _____ [1]

- (b) percentage profit made.

Answer (b) _____ [2]

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- 9 In the diagram ABCDE is a trapezium, $AC = 9$ cm, $DE = 4$ cm and $AE = 3$ cm. Lines AC and ED are parallel to each other.

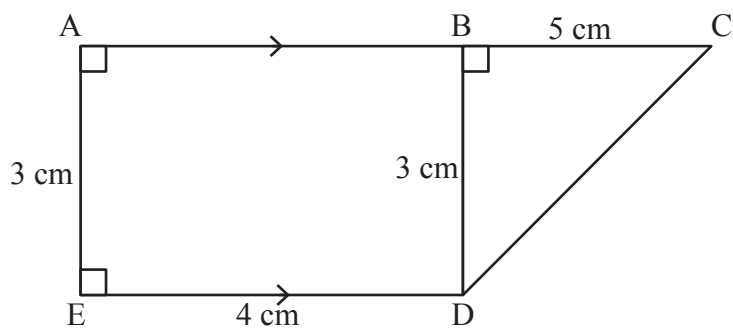


Fig. 9

Calculate the area of the trapezium.

Answer

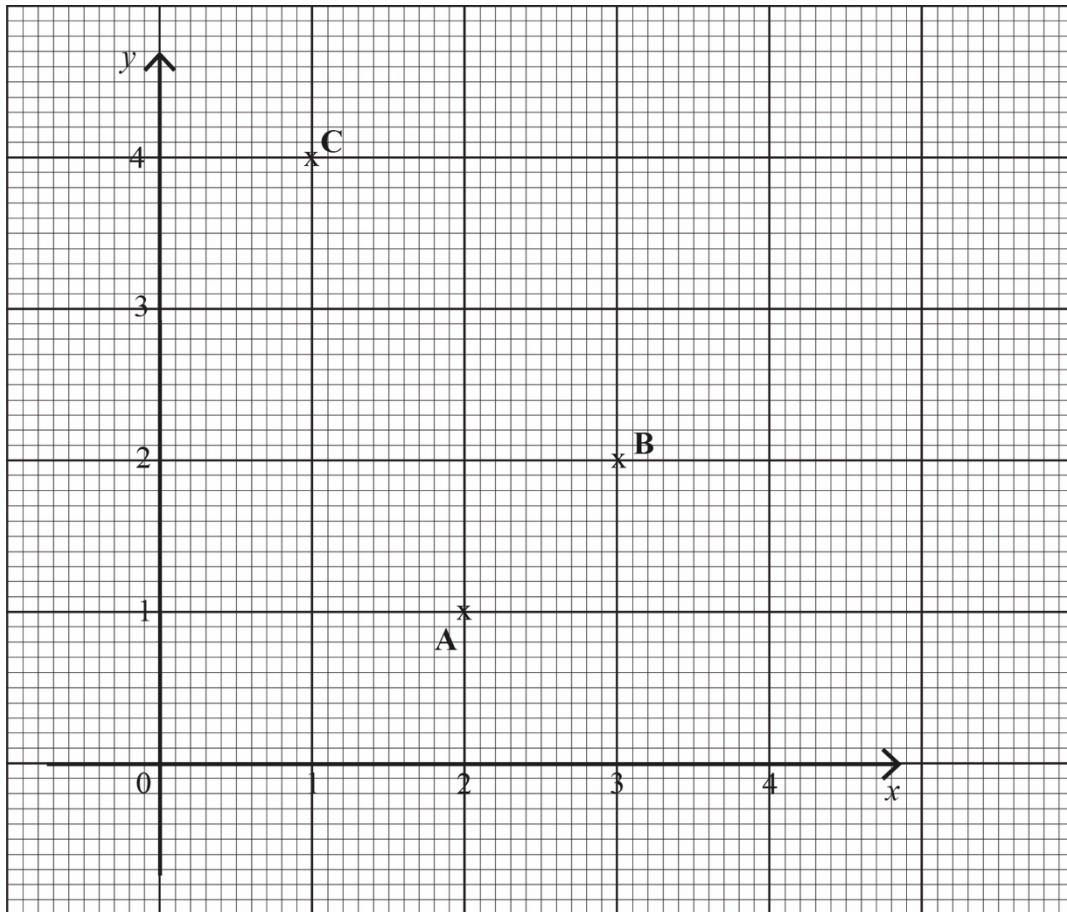
[3]

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- 10 The graph below shows a cartesian plane with points A, B and C plotted.



- (a) State the coordinates of B and C.

Answer (a) B()
C()

[2]

- (b) On the graph, mark the point D such that quadrilateral ABCD is a rectangle.

Answer (b) On the diagram

[1]

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11 Solve the equation

$$3x - 4(10 + 2x) = 0$$

Answer: _____ [3]

12 Factorise

(a) $9p - 3,$

Answer (a) _____ [1]

(b) $x^2 - x - 6.$

Answer (b) _____ [2]

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- 13** The data below shows the number of passengers in each bus leaving a bus terminus.

15; 18; 15; 15; 18; 13; 20; 9; 9; 15

The data is illustrated in the table below.

No of passengers	9	13	15	18	20
No of buses	2	x	y	2	1

- (a)** Find the number represented by

(i) x ,

Answer (a)(i) $x =$ _____ [1]

(ii) y .

Answer (ii) $y =$ _____ [1]

- (b)** State the name given to this table.

Answer (b) Name: _____ [1]

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- 14** Given the line **AB** in the answer space,
construct, using ruler and compasses only,
(a) the perpendicular bisector of the line **AB**,

Answer (a) On the diagram

[1]

- (b)** an angle of 60° at **A**.



Answer (b) on diagram

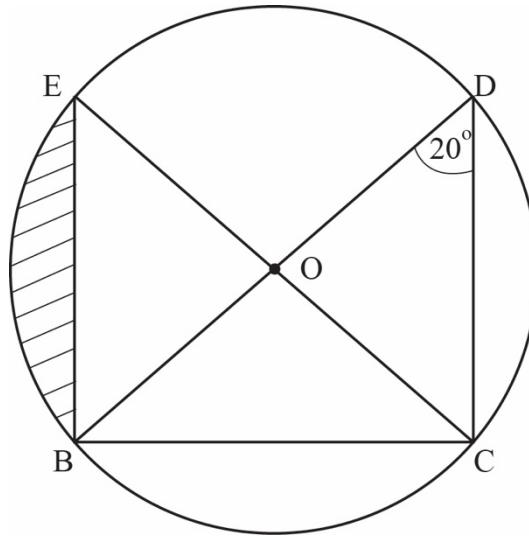
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- 15 The points B, C, D and E are on the circumference of a circle with centre O. $\widehat{BEC}$ and $\widehat{BDC}$ are subtended by the arc BC. $\widehat{BDC} = 20^\circ$. Lines BOD and COE are diameters of the circle.



- (a) Name the shaded region.

Answer (a) _____ [1]

- (b) Find

- (i) $\widehat{BEC}$,

Answer (b)(i) _____ [1]

- (ii) $\widehat{BOC}$.

Answer (ii) _____ [1]

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- 16** An employment agent gave a selection test to 9 job seekers. The marks obtained are shown in the table below.

Mark (x)	$30 \leq x < 40$	$40 \leq x < 50$	$50 \leq x < 60$	$60 \leq x < 70$	$70 \leq x < 80$
Frequency	2	1	4	1	1

Calculate an estimate of the mean mark.

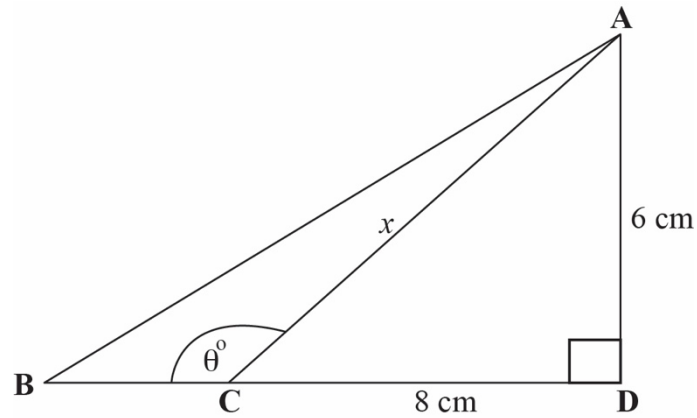
Answer: _____ [3]

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The diagram shows triangle ABD, in which $AD = 6$ cm, $CD = 8$ cm and BCD is a straight line. $\widehat{ADB}$ is a right angle and $\widehat{BCA} = \theta$.

- (a) Find the value of x .

Answer (a) _____ [2]

- (b) Find the value of $\tan \theta$ as a fraction in its simplest form.

Answer (b) _____ [1]

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18 Matrix $A = \begin{pmatrix} 0 & -3 \\ 4 & 3 \end{pmatrix}$.

Find the

(a) determinant of matrix A,

Answer: **(a)** _____ [1]

(b) inverse of matrix A.

Answer **(b)** _____ [2]

19 There are 8 identical apples in a bag except for colour 5 are green apples and 3 are red apples. One learner picks 2 apples from the bag, one at a time with replacement.

Find the probability that one is green and the other is red.

Answer: _____ [3]

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- 20** Given that $\vec{p} = \begin{pmatrix} 4 \\ -5 \end{pmatrix}$ and $\vec{q} = \begin{pmatrix} 2 \\ 6 \end{pmatrix}$.

Find $3\vec{p} - \frac{1}{2}\vec{q}$.

Answer: _____ [3]

- 21** The distance (D) that a car travels at a constant speed, varies directly as the time, T, taken. The car travels 300 km in 4 hours.

Find the

- (a)** law connecting D and T,

Answer (a) _____ [2]

- (b)** value of D when T = 3 hours.

Answer (b) _____ [1]

Answer (a) _____ [1]

Answer (b) _____ [2]

Answer (a) $x =$ _____ [1]

Answer (b) _____ [2]

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- 24** Mrs Moyo went to the market to buy fruits for her family. She bought both bananas and apples.

- (a)** If she bought x bananas and y apples, form another inequality other than $y > 0$.

Answer (a) _____ [1]

- (c)** She bought more bananas than apples.

Write down an inequality to represent this information.

Answer (b) _____ [1]

- (c)** Bananas cost 10 cents each and apples 25 cents each. Mrs Moyo wanted to use at most \$10.

Form an inequality for this information.

Answer (c) _____ [2]

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25 Two similar solid figures have volumes of 120 cm^3 and 360 cm^3 .

(a) Find the ratio of their

(i) sides,

Answer (a)(i) _____ [1]

(ii) areas.

Answer (ii) _____ [1]

(b) Given that the surface area of the solid of volume 120 cm^3 is 75 cm^2 , find the surface area of the solid of volume 360 cm^3 .

Answer (b) _____ [2]

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26 (a) Evaluate -2^3 .

Answer (a) _____ [1]

(b) Simplify $\left(\frac{8}{27}\right)^{\frac{2}{3}}$.

Answer (b) _____ [3]

27 (a) In a Mathematics class of 50 learners, 30 learners passed their end of term test.

Calculate the percentage of the learners who failed the test?

Answer (a) _____ [2]

(b) Simplify $\frac{5x}{4} - \frac{x-2}{3}$.

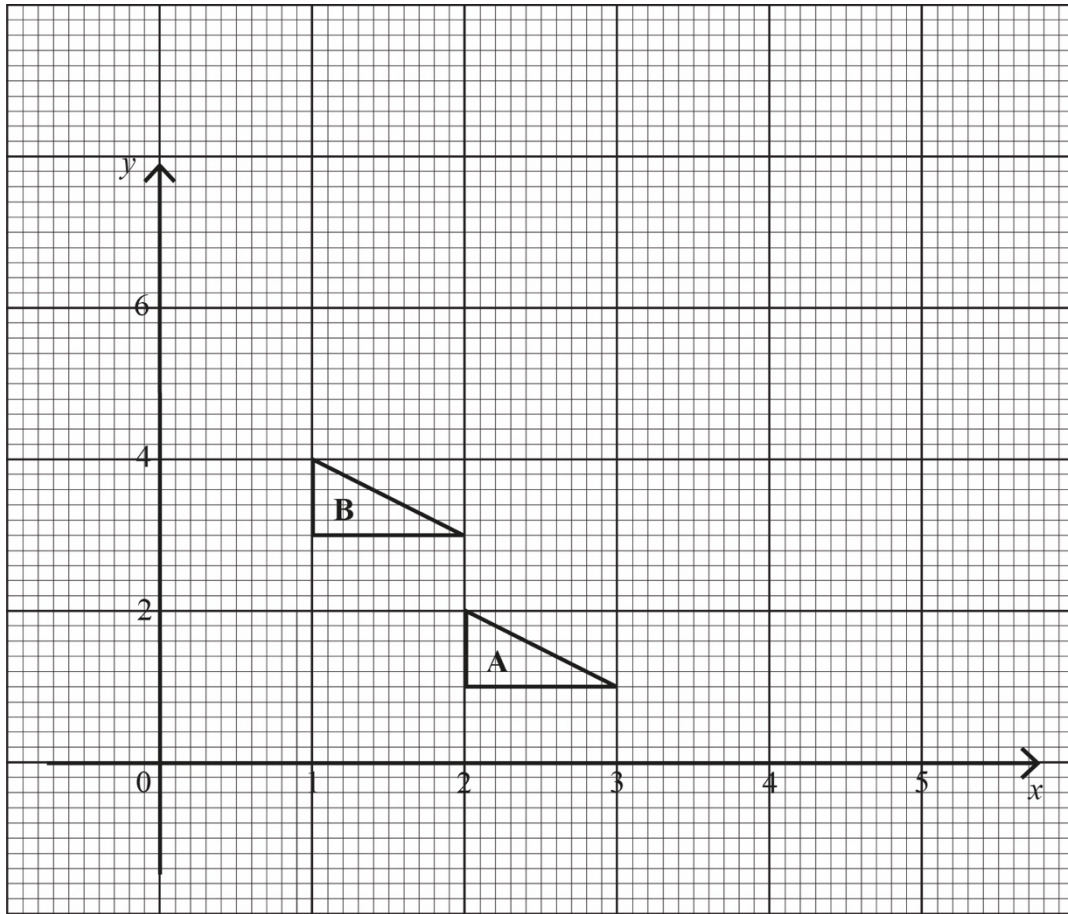
Answer (b) _____ [2]

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The graph shows two triangles A and B drawn on scale 2 cm to represent 1 unit on the x -axis and 2 cm to represent 2 units on the y -axis.

- (a) Describe fully the transformation that maps triangle A onto triangle B.

Answer (a) _____ [3]

- (b) Draw the image of triangle A after a translation with vector $\begin{pmatrix} 2 \\ -1 \end{pmatrix}$.

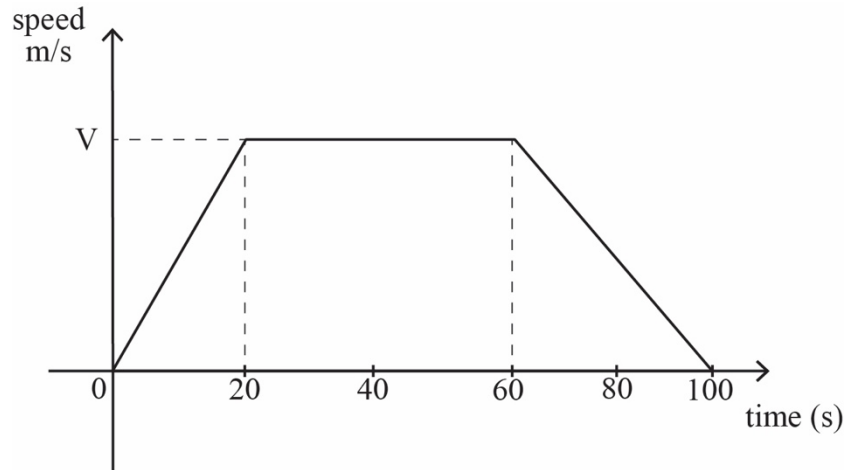
Answer (b) _____ [2]

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- 29** The diagram below is a speed-time graph for an athlete who took 100 seconds to complete a race. The athlete ran 1,75 km in 100 seconds.



Calculate the

- (a)** value of V .

Answer (a) _____ [3]

- (b)** acceleration of the athlete from time 60 seconds to time 100 seconds.

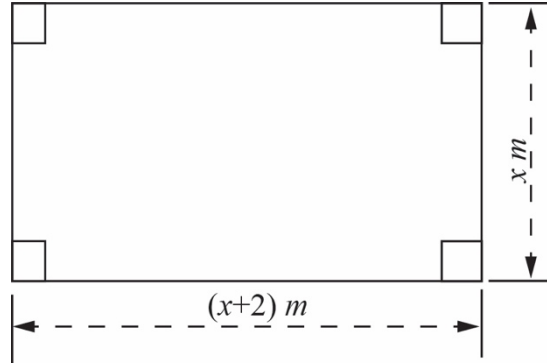
Answer (b) _____ [2]

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- 30** The diagram shows a rectangular piece of cloth measuring $(x + 2)$ metres by x metres. The area of the piece of cloth is 15 m^2 .



- (a)** Form an equation in x and show that it reduces to $x^2 + 2x - 15 = 0$.

Answer (a) _____ [2]

- (b)** Solve the equation $x^2 + 2x - 15 = 0$.

Answer (b) _____ [2]

- (d)** Hence or otherwise, find the length of the piece of cloth.

Answer (c) _____ [1]